

Performance Analysis of a Delay Constrained Data Offloading Scheme in an Integrated Cloud-Fog-Edge Computing System

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Abstract—The recent growth in intensive services and applications demand has triggered the functional integration of cloud computing with edge computing capabilities. One of the main goals is to allow a fast processing to tasks with strict real time constraints in order to lower the task dropping probability due to expiration of the associated deadlines. This paper deals with the performance evaluation and optimization of a three layers cloud-fog-edge computing infrastructure by resorting to the use of queueing theory results. In particular, a Markov queueing system model with reneging is proposed for the cloud subsystem, in order to consider the premature computation requests departure due to their deadline expiration. Furthermore, a computational resources allocation method is proposed with the aim at maximizing the social welfare metric, constrained to specific quality of service requirements. Finally, the proposed queueing theory analysis as well as of the computational resources allocation approach is validated by comparing the obtained analytical predictions with simulation results.

Index Terms—Mobile computing, offloading, queueing theory, performance optimization.

I. INTRODUCTION

DURING last decades, the cloud computing [1] architecture has held an undisputed dominant role in the network computing paradigms scenario, providing massive processing and storage capacity to the users. However, over the years, the large scale diffusion of devices always getting smarter, able to exchange data information to each other and with the surrounding environment, has led to a deep need to redesign the networks architectures. More in depth, the ever increasing presence of smart devices in our daily life has opened the doors to novel ubiquitous communication paradigms. In such a dynamic context, the Internet of Things (IoT) is emerged to express a wide reality consisting of heterogeneous smart devices generating big volume of data traffic and applications demand. Other typical aspects of the IoT may be represented by the mobility of the devices, which poses new challenges as concerns the seamless service continuity, or real-time execution constraints required for some classes of applications, such as augmented reality,

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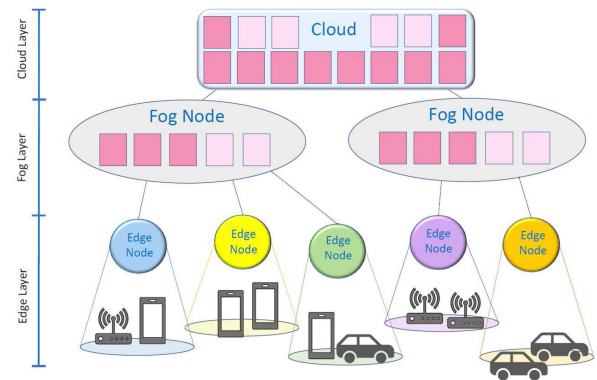


Fig. 1. Edge-fog-cloud Network Model.

health care applications or recognition assistance [2]–[5]. All these new challenges have triggered the tendency to migrate towards novel solutions, typically based on the deployment of computational nodes, characterized by a contained processing ability and storage supply, to the edges of the network, in order to reduce the network response latency [6]. The novel network paradigms based on this approach are named **edge computing (EC)** [7] and **fog computing (FC)**, and are able to provide low latency response and service continuity to mobile users.

However, in comparison to the existing public cloud based solutions, e.g., Microsoft Azur and Amazon AWS, the processing capacity of EC solutions is limited [8]. In order to address both the cloud and the EC/FC issues, the functional integration of these two approaches in a same computing infrastructure has recently gained momentum [2]. This solution enables massive improvements in system performance and users quality of service (QoS), introducing higher levels of flexibility for rapid computations and high mobility patterns. In this regards, the service providers point of view has to be considered, in order to provide a proper system resources exploitation. Towards this direction, the paper analyzes the performance of an integrated cloud-fog-edge computing infrastructure depicted in Fig. 1. At the Edge layer we have sets of **EC nodes (ECNs)** able to offer computation to the underlying devices. Each set of ECNs is linked to a fog computing node (FCN) belonging to the Fog layer. Finally, all the FCNs are linked to the cloud [9], [10]. ECNs and FCNs are modeled as Markov queueing systems with **finite capacity**. The entering of a computation request into the ECN or FCN is ruled by the number of requests waiting for computation

at its arrival instant, in relation to specific deadline constraints. Conversely, the cloud does not reject any computation request. As a consequence, we have resorted here to a Markov queueing system model with reneging, to take into account the premature departure of a request due to the expiration of the associated deadline.

Summarizing, the main contributions of this paper are

- Queueing theory analysis of the considered integrated cloud-fog-edge computing infrastructure;
- Performance optimization on the basis of the social welfare metric [11]–[13], constrained on a target referred to the dropping probability, i.e., the probability of the occurrence of the deadline expiration;
- Validation of the proposed analytical model and obtained analytical predictions throughout comparisons with numerical results derived by performing extensive computer simulation runs under realistic world conditions.

The rest of the paper is organized as follows. Section II provides an in-depth review of the related literature. Section III deals with the proposed system analysis based on the queueing theory models. In Section IV the problem formulation is addressed while Section V provides comparisons between the obtained analytical predictions with numerical results derived by resorting to extensive computer simulation runs assuming realistic word conditions. Finally, the conclusions are drawn in Section VI.

II. RELATED WORKS

Nowadays, due to the massive number of heterogeneous EUs devices running different and advanced applications, a big challenge is to properly handle computational resources in order to host all service requests with satisfactory performance. In accordance to this, several approaches and methodologies have been proposed in the scientific literature. In particular, a proper computing and networking resource allocation is the objective of [14], where a deep reinforcement learning based on the Q-learning algorithm is proposed in order to minimize the average service time based on network conditions. The computation offloading problem is addressed also in [15]. In this case, the authors analyze the performance of a blockchain-empowered mobile edge computing scheme taking into account both the mining and data processing tasks. In [15], the authors propose to combine deep reinforcement learning with a genetic algorithm approach to speed up the convergence time of the deep learning strategy. A deep reinforcement learning framework is adopted also in [16], in which the user equipments offload traffic on a vehicle network arranged to support users computation. Furthermore, the optimal policies for dynamic computation offloading are provided on the basis of a semi Markov process, considering stochastic mobility vehicle traffic models, and the maximization of the long-term utility of the computing network. Vehicular edge computing networks are studied also in [17], where the workload offloading and tasks computation scheduling is analyzed, supposing network nodes with high mobility levels. The paper [17] takes into account both the communication and computation resources by resorting to a Lagrangian relaxation problem. A collaborative approach

between cloud and vehicles to host users computational offloading is the subject of [18], in which a joint optimization of both the computational offloading policies and resource allocation is formulated, and a game theoretic approach is pursued. Within the optimal resource allocation in EC, authors in [19] design two solutions to optimally partitioning the edge servers resource blocks, aiming at minimizing the average service response time of IoT applications having strict service delay constraints. Moreover, computational offloading has been widely investigated also in [20], in which authors consider ultradense IoT networks and propose a combined game-theoretic greedy approach to minimize the overall system overhead, i.e., processing time and energy use, within the mobile computation offloading problem. Similarly, authors in [21] focus on the cloudlets offloading by performing virtual machine migration and power control approaches throughout the particle swarm optimization algorithm, aiming at simultaneously taking into account the computation, migration and transmission costs of the offloading procedure. Virtual machine migration and transmission power control are also applied in [22], in which the main goal is the minimization of the average user service delay. More in details, in [22], the service delay is given by the sum of both the processing and transmission delay terms. For this reason, the virtual machine migration is proposed to lower the delay due to the processing overhead, while transmission power control is proposed to reduce the transmission delay. Within this context, a heuristic in which the two approaches are integrated is proposed. Transmission power control is also adopted in [23], in which the mean field game theory is applied to design an offloading framework to manage both interference and handoffs in 5G ultra-dense small cell networks. Furthermore, some other works needing mention are given by [24]–[27]. In particular, the paper [24] designs a multi-layer edge computing framework to optimally allocate the tasks considering the trade-off between the computing and communication resources. A heterogeneous architecture has been also considered in paper [25], in which a joint assignment of task, computation capabilities and transmission resources is performed in order to minimize the latency of the whole system. Similarly, authors in [26] address the same joint optimization problem, considering the distinction between the wireless and the wired channels, connecting the edge servers with the devices and the cloud center with the edge servers, respectively. Also in this case, paper [26] aims at minimizing the network latency, by resorting to a min-max problem. A multi-objectives offloading framework is proposed also in [27], where authors propose a graph coloring based strategy to perform the physical resource blocks allocation, while the offloading decisions are performed considering the computation overhead estimated by all user equipment by the mobile edge servers, respectively.

In addition to the above methodologies, queueing theory analysis has recently received many attentions within the field of the computational networks. As a consequence, several papers dealing with different aspects of these systems are available from the literature. In particular, in [28] the authors focus on the optimization of the number of processors in a real time computing system, where arrivals are bursty and divided into two priority job classes. The two priority service requests are managed by setting a fixed number of processors to serve the

high priority queue, while the remaining servers are allocated on the basis of the waiting time on both the queues. The presence of two classes of customers has been analyzed also in [29], in which a system $M/M/1/K$ with weighted fair queueing discipline is studied. The conditions and performance of a computing system with finite input source is analyzed in [30]. Furthermore, in [30] a $M/M/C/m$ queueing system model is considered to derive the optimal number of processors through the fuzzy logic approach, aiming at minimizing the system maintenance cost. A novel method to assess the suitable number of servers in a queueing system with finite capacity is proposed in [31], on the basis of the level of customers satisfaction. In particular, the server optimization is performed here by considering three different metrics consisting of the system cost, its acceptability and the servers utilization rates. Authors in [32] optimize the number of processors on the basis of a $M/G/\infty$ system, focusing on the maximum exploitation of the production line, by considering a provider perspective. The study of the economic aspects of different versions of the computational systems have recently emerged in many papers. An example is represented by [13] where a dynamic control problem in an open Jackson network with limited capacity is formulated, in which the aim is to determine the suitable admission price to maximize the long term social welfare system metric. Paper [33] formalizes the cloud provider maximization profit, in which both service charges and business costs are considered in the system optimization.

With the emergence of the new network paradigms, queueing theory has been extensively applied to provide stochastic traffic analysis of next generation networks. Paper [34] aims at ruling offloading considering an heterogeneous networks scenario. The paper models both the partial and the full offloading policies, via Wi-Fi and cellular networks, considering reneging and service interruptions. The main aim of the paper is the optimal tradeoff between energy efficiency and system performance, and the heterogeneous offloading interfaces are represented through on/off Markov chain models. The offloading problem is investigated also in paper [35] authors use an on/off alternating renewals process which is analyzed to derive transmission delay and offloading efficiency. Then, a model with balking is proposed taking into account the WLAN status, the number of packets waiting for transmission and the associated deadline. Differently, an alternating renewal process is also used in paper [36], in order to model the availability of the WiFi network for the offloading strategy. The paper proposes a complete theoretical queueing analysis, and authors especially focus on a novel user patience metric. The customers impatience is analyzed also in paper [37], where a multi-server retrial queueing system is modeled, where customers may leave the system for balking or impatience. Furthermore, a more realistic customer behavior is formulated by using three parameters to weight probabilities. The offloading scheme in a hybrid cloud-fog computing system, for time critical application, is proposed in [38]. Paper [38] aims at minimizing the mobile devices power consumption, considering strict restrictions on the system response time. The paper [2] still considers a hybrid fog-cloud network as computational support in an IoT scenario. The paper addresses the healthcare services deployment, modeling each edge node as a $M/M/C$ system, the public cloud with a $M/M/\infty$ queue, and the private

cloud with a $M/M/c/K$ system. In [39] the mobile device resource sharing framework in a cloudlets network is proposed, to model offloading. The proposed framework consists of a $M/M/c/K$ queue system, and the optimal average service time for cloudlet is provided. This paper aims at investigating the performance of the integrated cloud-fog-edge computing infrastructure depicted in Fig. 1 by resorting to the use of suitable Markov queueing system models for the ECNs, FCNs and cloud, respectively. In addition to this, a performance optimization analysis is proposed on the basis of the social welfare metric [11]–[13], constrained on target dropping probability values. In particular, by means of the proposed heuristic strategy, the suitable number of active processors to be allocated at each ECN, FCN and cloud is derived in order to maximize the derived social welfare metric.

III. SYSTEM MODEL

A. Reference Scenario

We refer here to the three layers computing infrastructure depicted in Fig. 1. The EC layer consists of a suitable number of ECNs, each of them located at a given Base Station (BS) of a high speed, high reliable, low latency fifth generation (5G) wireless network. Each ECN provides computation services to all mobile users within its service area, i.e., within the coverage area of the related BS of the 5G network. FCNs belonging to the Fog layer, can be connected to a given number of ECNs by means of switched high speed links. FCNs provide tasks computation service to mobile users within the service area of the connected ECNs on the basis of the suitable procedure described later. Finally, we have the cloud layer where a cloud infrastructure connected by switched high speed links with all the FCNs of the Fog layer can provide computing service to all mobile users within all the ECN service areas on the basis of a suitable offloading procedure.

More in detail, we focus on a reference scenario where we have a set of tasks from mobile users requiring computation, and an integrated computing infrastructure composed of a set $\mathcal{E}$ of ECNs, several FCNs $\mathcal{F}$ belonging to the set $\mathcal{F}_c$, and a cloud. Each FCN offers computation support to a subset of ECNs $\mathcal{E}_F$ and, similarly, the remote cloud is devoted to provide support to $\mathcal{F}_c$. Hereafter we refer to the service area of an ECN as the geographical area within which a task originated in that area can be offloaded on the corresponding ECN throughout wireless links with a negligible latency. As stated before, we assume ECNs located at the BS sites of the 5G cellular network which supports, in conjunction with a high speed low latency core network, mobile users communication connections with the integrated computing infrastructure under consideration. As mobile users move across different cells during the time needed to complete a task computation, a handover procedure can be performed between neighboring cells to guarantee a seamless connection with the integrated computing infrastructure.

In this paper, due to the complexity and time consume of a task computation migration from one computation site to a neighboring one, we have considered here the communication plane (i.e., handover) separated by the computing plane (i.e., tasks computation). This means that, whenever a mobile user is no longer in the coverage of a given BS, it starts a handover

procedure but its offloaded task continues to receive computation service from the original ECN, even if the mobile user is not directly linked to the original ECN anymore. This is so, because a mobile user can be linked with the remote ECN by means of any BS to which the mobile user is connected that acts as a relay node.

The arrival processes of the task computation requests at the ECNs are assumed as independent and identically distributed Poisson processes with same mean arrival rate Λ .

At a first instance, a mobile user within a given EC service area submits a task computation request to the related ECN. This new arrived request is accepted if, on the basis of the task computation requests already in the ECN, it results, with suitable probability, that it can complete computation before expiration of its time deadline. Otherwise, the BS associated to the ECN redirects the task computation request to the linked FCN. Here again, the task computation request is accepted if, on the basis of the number of task computation requests already in the FCN, it is possible to guarantee with a suitable statistical uncertainty that its computation is completed within its time deadline. Differently, it is redirect through the core network to the cloud. Each new arrived task computation request at the cloud is accepted, hence, in this case, we can have task computations reneging due to the expiration of the tasks deadlines.

In performing our analysis we assume the task computation time at the ECNs, FCNs and cloud of the integrated computing infrastructure exponentially distributed with appropriate mean value (related to the different computation capabilities of each site). We will validate the goodness of this assumption by comparing the obtained analytical predictions with simulation results derived by assuming realistic world task computation time distributions. In addition to this, we consider that each task computation request has associated a deadline which expires after a time exponentially distributed [35], [40], [41], with mean value dependent on the specific computation site (EN, FN or cloud), in order to take into account the impact of the resulting different communication delays.

As a consequence, on the basis of our assumption, we have resorted to a M/M/K/H queueing system to model each ECN behavior, where K denotes the maximum number of task computations accepted by each ECN and H the number of available CPUs at each ECN. Similarly, the FCNs have been modeled as independent M/M/F/A queueing system. Finally, the cloud, being in this case the rejection of task computation request not allowed, has been modeled as a M/M/S queueing system. It is important to note that parameters K, H, F, A, S have to be derived on the basis of a suitable optimization approach in relation to specific QoS requirements in terms of task reneging probability at the cloud site less than a target value. It is important to note that at both the ECNs and FCNs, the task computation completion is assumed guaranteed, since any new arrived task computation request is admitted if this is assured with a suitable probability, as detailed later.

B. Edge Subsystem Analysis

On the basis of our previous assumptions, we have that each ECN belonging to $\mathcal{E}$ can be modeled as an independent

M/M/K/H queue system where task computation requests are served in their order of arrival (FCFS scheduling) and with each of them requesting an exponential service time with mean value T_E . Moreover, in this case, we have that the at most H task computation requests can be in the ECN at a time. This term is dependent on K in relation to a specific QoS requirement. Therefore, H is the highest value for which the resulting probability $P_{EOUT}(K, H)$ that the task computation request entering the system while there are $H - 1$ requests, waiting for service completion, does not complete its service before expiration of its deadline, is less than a target value, $P_{EOUT, tg}$.

Hence, given K and P_{EOUT} , the maximum number H of task computation requests accepted by each ECN is derived under the assumption (worst case) that each task computation request accepted by ECN completes service within its deadline. It follows that the time needed to a task computation request arrived while in the ECN there are $H - 1$ requests to complete its service, i.e., T , can be defined as the sum of $k = \lfloor \frac{H}{K} \rfloor + 1$ independent exponentially distributed random variables with mean values $\alpha = \frac{T_E}{K}$, and an independent exponentially distributed random variable with mean values $\beta = T_E$.

Therefore, the pdf of T can be obtained throughout the convolution of an Erlang distribution generating the random variable corresponding to the sum of the k independent exponential random variables previously introduced, and the exponential distribution with mean values β . After some algebraic manipulations, the corresponding pdf results to be

$$f_T(t) = \frac{\alpha^k \beta}{(k-1)!} e^{-\alpha t} \int_0^t (t-\tau)^{(k-1)} e^{-(\alpha-\beta)\tau} d\tau. \quad (1)$$

Hence, let E be the random variable exponentially distributed with mean value $\frac{1}{\mu_D}$ referred to the time deadline of the task computation request entering the ECN system in the H -th position, we have that $P_{EOUT}(K, H)$ is

$$\begin{aligned} P_{EOUT} &= P\{T > E\} = 1 - P\{T \leq E\} \\ &= 1 - \int_0^\infty \left(\int_\tau^\infty \mu_D e^{-\mu_D \sigma} d\sigma \right) f_T(\tau) d\tau \\ &= 1 - \int_0^\infty e^{-\mu_D \tau} f_T(\tau) d\tau. \end{aligned} \quad (2)$$

Consequently, parameters K, H have to be defined in order to have

$$P_{EOUT}(K, H) \leq P_{EOUT, tg} \quad (3)$$

Once K, H have been defined to satisfy (3), any new task computation request arrived while the ECN is in state H is redirect to the linked FN for a successive consideration. By referring to the state diagram of the M/M/K/H queue system under consideration, shown in Fig. 2, we have

$$\lambda_n = \begin{cases} \Lambda, & 0 \leq n < H \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

and

$$\mu_n = \begin{cases} n\mu_s, & 1 \leq n < K \\ K\mu_s, & K \leq n \leq H, \end{cases} \quad (5)$$

$P_{EOUT} \leq P_{EOUT, tg}$
این احتمال آن است که وظیفه ای تازه وارد شده و قبل از آن H-1 تا وظیفه دیگر نیز وجود دارد، نتواند پردازشش را قبل از تمام شدن مهلت از زمان اش تکمیل کند.
مقدار T بیانگر میزان زمان لازم برای وظیفه تازه رسیده است تا بخواهد پردازشش تمام شود. این زمان باید کمتر از مهلت وظیفه باشد این مان برابر با کف تقسیم H بر K به علاوه یک است. این رو این مدت زمان را با متغیر های تصادفی یک تا K در نظر میگیرد.

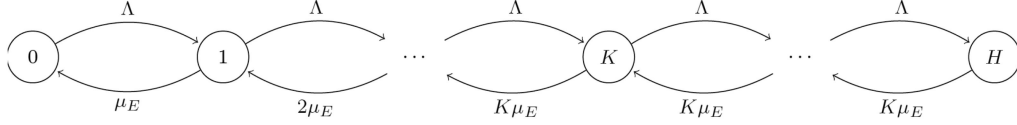


Fig. 2. ECN Markov chain.

where $\mu_s = 1/T_s$ and the system state represented by the number n is the task computation requests in the system.

Therefore, p_n , i.e., the probability of having n task computation requests in the ECN, for $1 \leq n < K$, results to be

$$p_n = p_{0,E} \frac{\Lambda^n}{n! \mu_s^n}, \quad (6)$$

while for $K < n \leq H$ is

$$p_n = p_{0,E} \frac{\Lambda^n}{K^{n-K} K! (\mu_s)^n}. \quad (7)$$

The term $p_{0,E}$, i.e., the probability of having no task computation requests in the ECN system, in (9) and (7) can be derived by imposing the validation of the state probability normalization condition $p_{0,E} + \sum_{n=1}^{\infty} p_n = 1$ according to [10] obtaining

$$p_{0,E} = \left[\sum_{n=0}^{K-1} \frac{\Lambda^n}{n! \mu_s^n} + \sum_{n=K}^H \frac{\Lambda^n}{K^{n-K} K! \mu_s^n} \right]^{-1}. \quad (8)$$

Moreover, it is easy to note that the task computation request blocking probability P_B equals to the probability of having the ECN system in state H , i.e., $P_B = p_H$.

Then, the mean time spent in the system by any task computation request results, through application of the Little's formula [10]

$$T_{p,E} = \frac{\sum_{x=0}^H x p_x}{\Lambda(1 - p_H)}, \quad (9)$$

where $\Lambda(1 - p_H)$ is the mean task computation requests arrival rate at each ECN system.

C. Fog Subsystem Analysis

According to our assumptions, we have here again that the FCNs can be modeled as independent M/M/F/A queue systems where task computation requests are served in their order of arrival (FCFS scheduling) and each task computation request requires an exponential service time with mean value T_F . As for the ECNs, we have that at most A task computation requests can be in each FCN at a time with this term dependent on F in relation to the specific QoS requirement previously introduced, i.e., $P_{FOUT}(F, A) \leq P_{FOUT,tg}$. Also in this case, given the F and P_{FOUT} values, the number A of task computation requests accepted by each FCN, according to the considered worst case

analysis, is the maximum value for which the following constraint is satisfied

$$P_{FOUT}(F, A) \leq P_{FOUT,tg}, \quad (10)$$

where now $P_{FOUT}(F, A)$ is

$$\begin{aligned} P_{FOUT} &= P\{U > E\} = 1 - P\{U \leq E\} \\ &= 1 - \int_0^{\infty} e^{-\mu_D \tau} f_U(\tau) d\tau, \end{aligned} \quad (11)$$

in which

$$f_U(t) = \frac{\pi^\theta \xi e^{-\pi t}}{(\theta - 1)!} \int_0^t (t - \tau)^{(\theta-1)} e^{(\pi-\xi)\tau} d\tau, \quad (12)$$

and, as in the edge subsystem case, we have $\pi = \frac{T_E}{F}$, $\xi = T_F$, and $\theta = \lfloor \frac{A}{F} \rfloor + 1$. In this case we have that any new task computation request arrived while the FCN is in state A is routed to the cloud. It is important to highlight that being each FCN linked with a number $\mathcal{Y}$ of ECNs, the task computation requests arrival process is Poisson with mean rate Γ , equals to $\mathcal{Y} \Lambda P_B$, resulting from the superposition of $\mathcal{Y}$ independent Poisson processes with equal mean rate ΛP_B [10]. The state diagram of the M/M/F/A queue system under consideration is shown in Fig. 4. The system state is again represented by the number n of task computation requests in the FN and the queue system parameters are given by

$$\lambda_f = \begin{cases} \Gamma, & \text{if } \exists \varepsilon \in \mathcal{E}_F \text{ s.t. } n = H \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

$$\mu_f = \begin{cases} f(\mu_F + \mu_D), & 1 \leq f < F \\ F\mu_F + f\mu_D, & l \geq A. \end{cases} \quad (14)$$

Hence, the state probability p_f of having f requests in the system, $1 \leq f < F$, is

$$p_f = p_{0,FN} \frac{\Gamma^f}{f! (\mu_F + \mu_D)^f}, \quad (15)$$

and, for $F \leq f < A$

$$p_f = p_{0,FN} \frac{\Gamma^f}{(F-1)! (\mu_F + \mu_D)^{F-1} \prod_{y=F}^f (y\mu_D + F\mu_F)}. \quad (16)$$

As before, the term $p_{0,FN}$, i.e., the probability of having no task computation requests in the considered FCN, in (15) and (16) can be derived by application of the state probabilities normalization condition as in (17), shown at the bottom of this page, with the

$$p_{0,FN} = \left[\sum_{f=0}^{F-1} \frac{\Gamma^f}{f! (\mu_D + \mu_F)^f} + \sum_{f=F}^A \frac{\Gamma^f}{(F-1)! (\mu_D + \mu_F)^{F-1} \prod_{y=F}^f (y\mu_D + F\mu_F)} \right]^{-1} \quad (17)$$

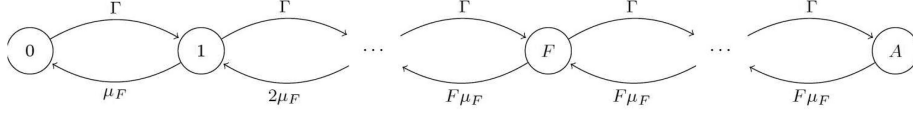


Fig. 3. FCN Markov chain.

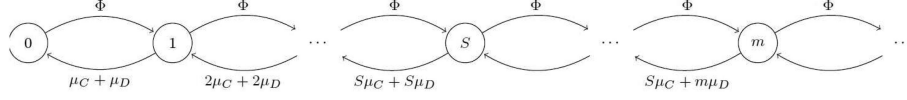


Fig. 4. Cloud Markov chain.

task computation requests blocking probability P_B , in this case, equals to the probability to have the considered FCN in state A , i.e., p_A .

As before, the mean time spent by each task computation request in a FCN through the Little's formula is

$$T_{p, FN} = \frac{\sum_{x=0}^{\infty} x p_x}{\Gamma(1 - p_A)}. \quad (18)$$

where $\Gamma(1 - p_A)$ is the mean task computation requests arrival rate.

D. Cloud Subsystem Analysis

In the cloud case, we have no limitations on the accepted tasks computation requests. Hence, it follows that the cloud can be modeled as a M/M/S queue system, with FCFS selection policy and S available CPUs. In this case, we have that the cloud may be linked to a number of $\mathcal{F}$ FNs. So that, according to our assumptions, we have that the task requests arrival process is Poisson with mean rate Φ equal to $\mathcal{F}\Lambda P_A$. The associated system state diagram is provided in Fig. 4 where the system state is considered as the number of task computation requests m in the cloud. In particular, we have

$$\lambda_m = \begin{cases} \Phi, & \text{if } \exists \eta \in \mathcal{F}_C \text{ s.t. } f = A \\ 0, & \text{otherwise} \end{cases} \quad (19)$$

$$\mu_m = \begin{cases} m(\mu_C + \mu_D), & 1 \leq m < S \\ S\mu_C + m\mu_D, & m \geq S. \end{cases} \quad (20)$$

Consequently, the state probability p_m , that is the probability of having m tasks in the cloud subsystem, for $1 \leq m < S$, is

$$p_m = p_{0,C} \frac{\Phi^m}{m!(\mu_C + \mu_D)^m}, \quad (21)$$

while for $m \geq S$, we have (22), shown at the bottom of this page. In (21) and (22) the term $p_{0,C}$, i.e., the probability of having no task computation requests in the cloud system, is given as in (23), shown at the bottom of this page, by imposing again the

validating of the state probability normalization condition for this case. Similarly to the previous analysis, the mean time spent by each task computation request in the cloud system is

$$T_{p,C} = \frac{\sum_{x=0}^{\infty} x p_x}{\Phi}. \quad (24)$$

Finally, in this case the reneging probability, i.e., the probability that a task computation request in the cloud does not complete its service due to its deadline expiration, P_D , can be obtained according to [42], as

$$P_D = T_{p,C} \mu_D. \quad (25)$$

IV. PROBLEM FORMULATION

The main aim of the paper is to pursue an optimization procedure in order derive the proper number of active processors allocated to each ECN, FCN and cloud, as well as the maximum number of computation requests accepted by each ECN and FCN, respectively. The optimization procedure is intended here as the maximization of the social welfare [11]–[13] function, through which both the users and provider points of view can be simultaneously taken into account, reaching a good trade-off between the parts involved in the proposed network infrastructure.

Furthermore, assuming Δ_{Eff} as the whole infrastructure rate of satisfied requests, the social welfare metric is defined as

$$\begin{aligned} \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S') \\ = U(\Delta_{Eff,E} + \Delta_{Eff,F} + \Delta_{Eff,C}) \\ - V(T_{p,E} + T_{p, FN} + T_{p,C}) \\ - \sum_{z=1}^K r b_z - \sum_{w=1}^F u c_w - \sum_{j=1}^S s d_j, \end{aligned} \quad (26)$$

where $\Delta_{Eff,E} = \Lambda(1 - p_H)$, $\Delta_{Eff, FN} = \Gamma(1 - p_a)$, which represent the mean values of the computation requests arrival

$$p_m = p_{0,C} \frac{\Phi^m}{(S-1)!(\mu_D + \mu_C)^{S-1} \prod_{g=S}^m [g\mu_D + S\mu_C]} \quad (22)$$

$$p_{0,C} = \left[\sum_{m=0}^{S-1} \frac{\Phi^m}{m!(\mu_D + \mu_C)^m} + \sum_{m=S}^{\infty} \frac{\Phi^m}{(S-1)!(\mu_D + \mu_C)^{S-1} \prod_{g=S}^m [g\mu_D + S\mu_C]} \right]^{-1} \quad (23)$$

rates accepted by each ECN and FCN for which we have assumed guaranteed the service completion. Likewise, $\Delta_{Eff,C} = \Phi(1 - P_D)$ is the mean rate of computation requests completing service without deadline expiration at the cloud site. In particular, Δ_{Eff} represents the sum of the single accepted requests flows deriving from the single subsystem, although Λ , Γ , and Φ are the arrivals rate of the edge node, fog node, and the cloud subsystems, respectively. Therefore, the mean number of computation requests accepted by the three computation subsystems are given by $\Delta_{Eff,E} = \Lambda(1 - p_H)$, $\Delta_{Eff,FN} = \Gamma(1 - p_a)$, and $\Delta_{Eff,C} = \Phi(1 - P_D)$ which define Δ_{Eff} as the sum of these three quantities, i.e., $\Delta_{Eff} = \Delta_{Eff,E} + \Delta_{Eff,FN} + \Delta_{Eff,C}$. It is important to highlight that (26) represents the gap between the revenue due to the users service accomplishments and the penalty associated to the time spent in the computation subsystems, minus the provider costs associated to different CPUs. Therefore, by considering this metric, we indirectly take into account the service provider and the users interests, since a higher number of users accomplishments implies greater service gains and, from the other side, an efficient resource exploitation, i.e., the CPUs activation policy, triggers a greater number satisfied users. Furthermore, K', H', F', A', S' represents the system parameters values for the integrated cloud-fog-edge computation infrastructure, that have to satisfy the following architectural constraint:

$$K' \leq K, \quad H' \leq H, \quad F' \leq F, \quad A' \leq A, \quad S' \leq S. \quad (27)$$

In addition to this, in (26) we have that

- U is the gain associated to the client service accomplishment while V is a penalty associated to each unit of time spent by a computation request in an ECN, FCN or cloud;
- Assuming the service provider operating cost associated to a CPU dependent on its location, i.e., ECN, FCN or cloud, r represents the service provider cost for CPU available at a ECN, u is the cost related to a CPU located at a FCN, while s is the cost associated to an active CPU at the cloud site;
- $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ are binary vectors whose elements b_i , with $1 \leq i \leq K$, c_i with $1 \leq i \leq F$, and d_i $1 \leq i \leq S$ equal to 1 if at the ECN, FCN or cloud site, respectively, the i -th CPU is active, or 0 otherwise.

Hence, being our goal that of maximizing the social welfare metric defined in (26), we can formulate our problem as

$$\max_{\mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S'} \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S'), \quad (28)$$

s.t.

$$P_{Deadline} \leq P_{D, target}, \quad (29)$$

$$K' \leq K, \quad (30)$$

$$H' \leq H, \quad (31)$$

$$F' \leq F, \quad (32)$$

$$A' \leq A, \quad (33)$$

$$\Phi < S\mu_C, \quad (34)$$

where

- Constraint (29) represents a QoS target constraint referred to each ECN, FCN and cloud site on the probability that a computation request does not complete its service due to its deadline expiration;
- Constraints (29)–(33) represent the architectural restricts of the integrated cloud-fog-edge computation infrastructure;
- Constraint (34) refers to the stability condition for the cloud subsystem, interpreted in its most strict form.

A. Proposed Heuristic

Due to the intrinsic difficulty of problem (28)–(29), this paper proposes an approximated heuristic to determine the suitable number of CPUs to be allocated at each computation site, i.e., ECNs, FCNs, and cloud, in addition to the maximum number of computation requests accepted at each ECN and FCN in order to guarantee a specific QoS in relation to the probability that a computation request does not complete its service at a given site due to its deadline expiration. The proposed social welfare maximization procedure is iterative and, for each computation site, acts as follow

- 1) Let S, F, K, H , and $A = 1$ be the maximum architectural capacity of the whole system. Start with $S' = 1, F' = 1, K' = 1, H' = 1$, and $A' = 1, \phi = 0, \xi = 0, \theta = 0, \pi = 0, \psi = 0$;
- 2) Compute P_D . If $P_D \leq P_{D, target}$ then terminate, otherwise evaluate
 - a) If $K' = K$, i.e., it cannot be incremented, set $\phi = 1$ and jump to b), otherwise evaluate $A_1 = \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S')$ with $S', K' = K' + 1, H', F', A'$;
 - b) If $H' = H$, i.e., it cannot be incremented, set $\xi = 1$ and jump to c), otherwise evaluate $A_2 = \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S')$ with $S', K', H' = H' + 1, F', A'$;
 - c) If $F' = F$, i.e., it cannot be incremented, set $\theta = 1$ and jump to d), otherwise evaluate $A_3 = \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S')$ with $S', K', H', F' = F' + 1, A'$;
 - d) If $A' = A$, i.e., it cannot be incremented, set $\pi = 1$ and jump to e), otherwise evaluate $A_4 = \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S')$ with S', K', H', F' , and $A' = A' + 1$;
 - e) If $S' = S$, i.e., it cannot be incremented, set $\psi = 1$ and jump to 3), otherwise evaluate $A_5 = \mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S')$ with $S' = S' + 1, K', H', F'$, and A' ;
- 3) If $(\psi \wedge \xi \wedge \theta \wedge \pi \wedge \phi)$ is equal to 1 terminate, otherwise select the resource allocation among A_1, A_2, A_3, A_4 , and A_5 which **minimizes** $\mathcal{F}(\Delta_{Eff}, \mathbf{b}, \mathbf{c}, \mathbf{d}, K', H', F', A', S')$;
- 4) Perform allocation in accordance with the previous step;
- 5) Repeat from step 2).

TABLE I
SYSTEM PARAMETER

Operating System	Ubuntu 16.04
Programming Platform	Matlab
CPU	Intel i7-3770 3.40Ghz
RAM	16 GB

V. NUMERICAL RESULTS

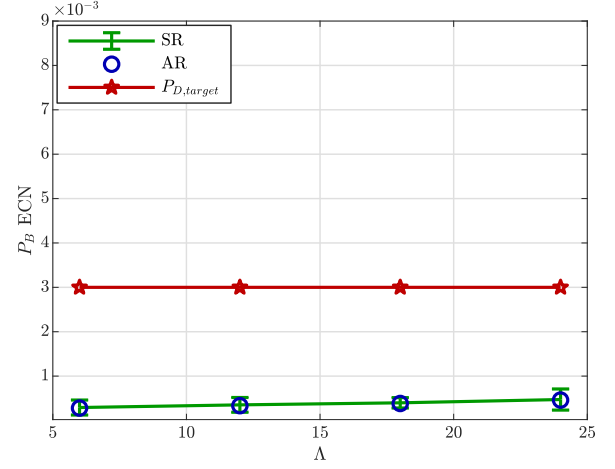
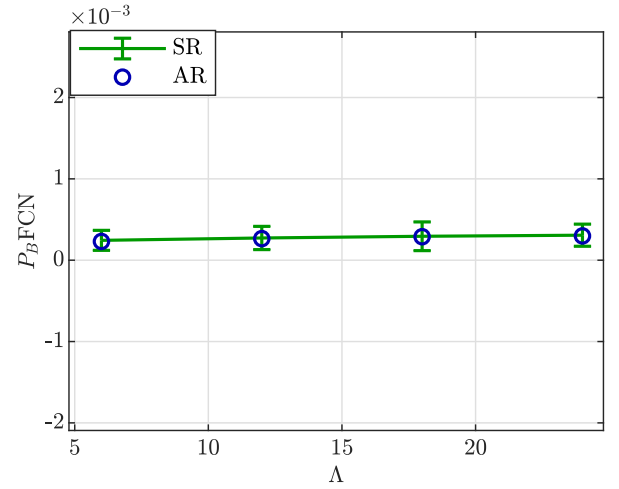
In order to validate the effectiveness of the proposed analytical approach, this section provides comparisons between the obtained analytical results (AR) based on the assumed Markov queueing system models with numerical results (SR), derived by resorting to extensive computer simulation runs by considering actual arrival and service time distributions under the assumption of same mean values [10]. Furthermore, where clearly visible, the confidence bars have been inserted in the corresponding plots. In performing our analysis, conducted on the basis of specifications reported in Table I, we assume the cloud system equipped with a maximum number of $S = 16$ CPUs, each FCN with a maximum number F of CPUs equal to 13 maximum, and each ECN with a maximum of $K = 10$ CPUs. Furthermore, we refer to an edge layer composed of 6 ECNs and a fog layer with 3 FCNs.

According to [43], in order to test the validity of the proposed theoretical model, we have analyzed here, under the assumption of equal mean values, the case of computation requests arrivals modeled as a Normal distribution, with service time following a hyperexponential distribution as in [10]. Furthermore, we have assumed that applications require an equal mean computation time at each computation site. In particular, in the cloud case, it results equal to 0.4 s, with a $\mu_C = 2.5 \text{ s}^{-1}$. Likewise, the mean requests computation time on each FCN is 0.2 s, with $\mu_F = 5 \text{ s}^{-1}$, while the mean requests computation time on each ECN has been assumed equals to 0.1 s, with $\mu_E = 10 \text{ s}^{-1}$. The system performance is measured in terms of computation requests completion failure probability at each ECN, FCN and cloud site, P_B limited to ECNs and FCNs, social welfare metric referred to the integrated cloud-fog-edge computing infrastructure under consideration.

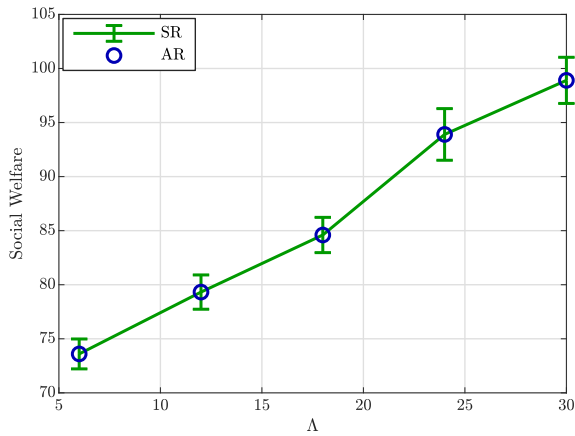
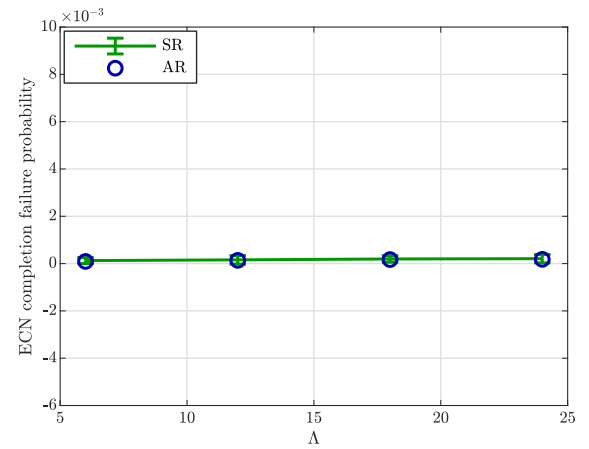
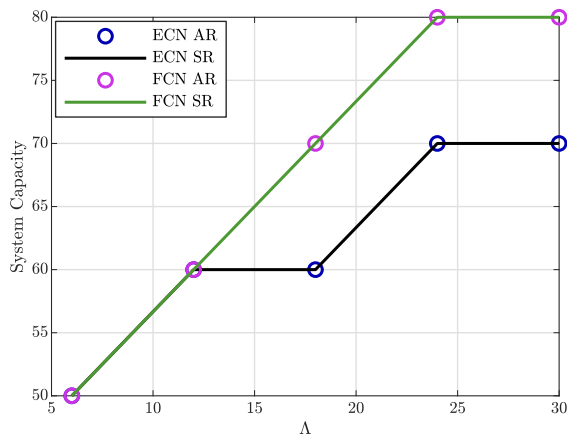
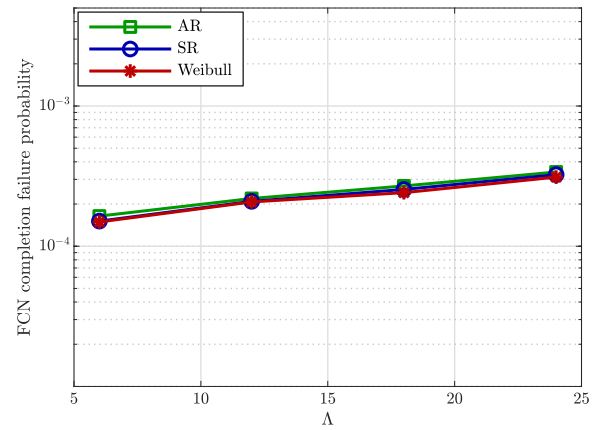
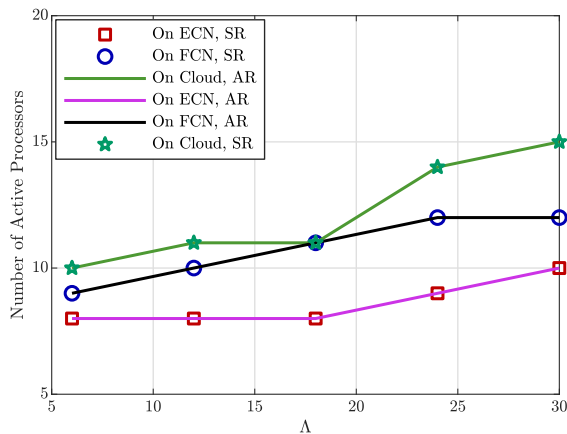
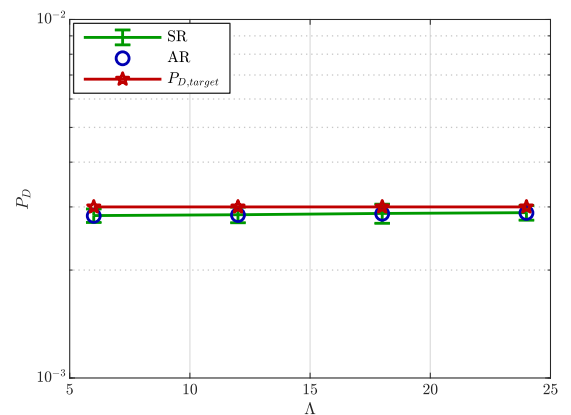
All the numerical results presented here are referred to the mean values obtained from 2000 independent runs of simulations. Fig. 5 depicts the behavior of P_B at an ECN as a function of Λ for H and K values derived in order to satisfy the constraint of a resulting $P_{EOUT}(K, H)$ less than 3.00×10^{-3} .

Likewise, Fig. 6 shows the behavior of P_B at each FCN as a function of Λ with the system parameters A and F derived in order to satisfy the constraint $P_{FOUT}(F, A) \leq 3.00 \times 10^{-3}$. It is important to stress that in all previous figures a very good agreement between analytical predictions based on the considered Markov queueing system models with the simulation results derived under real world arrival and service time distributions [10].

The results related to the proposed heuristic based on the maximization of the social welfare metric, defined in (26) with

Fig. 5. Computation requests blocking probability at each ECN as function of Λ .Fig. 6. Computation requests blocking probability at each FCN as function of Λ .

$V = 4$, $U = 15$ per time unit, $r = 20$, $u = 16$, and $s = 12$, are provided in Fig. 7 as a function of the parameter Λ . The values of the systems parameters, i.e., number of CPUs allocated at each ECN, FCN and cloud, maximum number of computing requests accepted by each ECN and FCN, that maximize (26), are shown in Figs. 8 and 9, respectively, as a function of Λ . Finally, in Figs. 10–12 is shown the resulting computation requests completion failure probability at each ECN, FCN and cloud site, respectively, as a function of Λ . The assumed computation requests completion failure probability target value (i.e., 3.00×10^{-3}) for all the computation sites is given in Fig. 10. In all these figures comparisons with the simulation results derived under proper assumptions for the computing requests arrival process and service time are also given in order to highlight the goodness of our Markov approach. Furthermore, it is important to note that the trends of H and A in Fig. 8, in relation to Figs. 9 and 7, reveal that the size of H and A increases when the cost of the activation of new processors is higher in comparison to the

Fig. 7. Social Welfare metric as function of Λ .Fig. 10. Computation requests completion failure probability at each ECN as function of Λ .Fig. 8. Maximum number of computation requests accepted by each ECN and FCN resulting by our heuristic as function of Λ .Fig. 11. Computation requests completion failure probability at each FCN as function of Λ .Fig. 9. Number CPUs to be allocated at each ECN, FCN and cloud resulting by our heuristic as function of Λ .Fig. 12. Computation requests completion failure probability at the cloud site as function of Λ .

increment of the size of H or A . It is important to stress that, the ever-increasing trend of the social welfare function shown in Fig. 7, demonstrates that the system parameters provided in Figs. 8 and 9 allow a good exploitation of system resources in reference to the set QoS target, on the basis of the service provision cost which characterized each subsystem. Finally, with

the aim at broadening our analysis, in Fig. 11, we have also considered the case of a Weibull distribution for the service times modeling [10]. Also in this case, the closeness between the AR and the results derived from simulation is clearly evident, confirming the validity of the proposed approach.

VI. CONCLUSION

This paper has investigated the behavior of an integrated cloud-fog-edge computing infrastructure by resorting to the use of suitable Markov queueing models. A system design procedure based on an heuristic method for the maximization of a social welfare metric, conveniently defined, has been also proposed in order to derive the most appropriate values for the considered system parameters, i.e., number of CPUs to be allocated at each ECN, FCN and cloud, and maximum number of computation requests accepted by each ECN and FCN, in order to satisfy specific QoS requirements in terms of probability that a computation request does not complete its service at a given site, i.e., ECN, FCN or cloud, due to its deadline expiration less than specific target values. Finally, extensive simulations results have been provided in order to confirm the validity of the proposed theoretical approach, by exhibiting a strong closeness with the obtained analytical predictions derived under the assumption of more realistic computing requests arrivals and service time distributions.

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